Synthetic Homotopy Theory

Exercises and Solutions for the

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NB Subject to change: Refresh if needed

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Chapter 1

Truncated and connected types

1.1 Exercises

1. Show that for $n \geq m$ and any type A we have

$$\|\|A\|_n\|_m \simeq \|A\|_m.$$

2. Show that a type X is n-connected if and only if, for every n-type Y, the diagonal map

$$\delta : Y \to (X \to Y), \quad y \mapsto_\delta (x \mapsto y),$$

is an equivalence.

3. Show that the map $|_|_n : X \to \|X\|_n$ is n-connected.

4. Consider a map $f : X \to Y$. We define its **n-image** to be the type $\mathrm{im}_n(f) := (y : Y) \times \|\mathrm{fib}_f(y)\|_n$.

   a) Show that we can factorise f through the n-image, that is, construct a commuting diagram:

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "p"'] & & Y \\ & \mathrm{im}_n(f) \arrow[ur, "i"'] & \end{tikzcd}$$

   b) Show that p is n-connected and i is n-truncated.

   c) Show that if Z is any type, $g : X \to Z$ is n-connected, and $h : Z \to Y$ is n-truncated with $f = h \circ g$, then there is a unique equivalence

$e : \mathrm{im}_n(f) \to Z$ with $e \circ p = g$ and $h \circ e = i$:

$$\begin{tikzcd}
 & Z \arrow[dd, dotted, no head] \arrow[dr, "h"] & \\
X \arrow[ur, "g"] \arrow[rr, "f"] \arrow[dr, "p"'] & & Y \\
 & \mathrm{im}_n(f) \arrow[ur, "i"'] &
\end{tikzcd}$$

(*Hint*) It's easier if you assume g and h are projections of dependent sums.

5. (Eckmann–Hilton). All paths will take place in a fixed type X.

   a) Given paths $p_0, p_1 : x = y$ and $q_0, q_1 : y = z$, and higher paths $r : p_0 = p_1$ and $s : q_0 = q_1$, define their **horizontal multiplication** $s * r : q_0 \cdot p_0 = q_1 \cdot p_1$ by path induction on r and p_0.

$$\begin{tikzcd}
x \arrow[r, bend left, "p_0"] \arrow[r, bend right, "p_1"'] \arrow[r, phantom, "\Downarrow r"] & y \arrow[r, bend left, "q_0"] \arrow[r, bend right, "q_1"'] \arrow[r, phantom, "\Downarrow s"] & z
\end{tikzcd}$$

   b) Given paths fitting into the following diagram

$$\begin{tikzcd}
x \arrow[r, bend left, "p_0"] \arrow[r, "p_1"'] \arrow[r, phantom, shift left=3, "r_0\Downarrow"] \arrow[r, phantom, shift right=3, "r_1\Downarrow"] & y \arrow[r, bend left, "q_0"] \arrow[r, "q_1"'] \arrow[r, phantom, shift left=3, "\Downarrow s_0"] \arrow[r, phantom, shift right=3, "\Downarrow s_1"] & z
\end{tikzcd}$$

   show we have the **interchange law**:

$$(s_1 * r_1) \cdot (s_0 * r_0) = (s_1 \cdot s_0) * (r_1 \cdot r_0)$$

   c) Deduce that for $n \geq 2$, the path composition in $\Omega^n(X, x)$ is commutative, and thus that $\pi_n(X, x)$ is abelian.

6. (The Fundamental Theorem of Identity Types)

   Let $R : X \to X \to \mathcal{U}$ be a reflexive binary relation. The following are equivalent

   a) The obvious map[1] $x =_A y \to R(x, y)$ is an equivalence.

   b) For any $x : X$, the total space $(y : X) \times (R(x, y))$ is contractible.

[1] i.e. the map defined by path induction which sends refl_x to the term of type $R(x, x)$ given to us by reflexivity of R

7. Recall that $\mathcal{U}^{\leq n} := (X : \mathcal{U}) \times \mathsf{is}\text{-}n\text{-}\mathsf{type}\ (X)$ is an $(n+1)$-type. This allows us to define $R : \|X\|_{n+1} \times \|X\|_{n+1} \to \mathcal{U}^{\leq n}$ by the truncation elimination:

$$R(|x|_{n+1}, |y|_{n+1}) := \|x = y\|_n$$

Apply the fundamental theorem of identity types with the above family to deduce that $(|x|_{n+1} =_{\|X\|_{n+1}} |y|_{n+1}) \simeq \|x = y\|_n$

8. Show that when X is connected, that $\pi_n(X, x_0) \cong \pi_n(X, x_1)$ are merely isomorphic for any $x_0, x_1 : X$, that is, construct an element of the type

$$(x_0, x_1 : X) \to \|\pi_n(X, x_0) \simeq \pi_n(X, x_1)\|.$$

9. We define the observational equality type family of the natural numbers $E : \mathbb{N} \to \mathbb{N} \to \mathcal{U}$ by

$$\begin{aligned} E(0,0) &= 1 \\ E(\mathrm{succ}(n), 0) &= 0 \\ E(0, \mathrm{succ}(n)) &= 0 \\ E(\mathrm{succ}(n), \mathrm{succ}(m)) &= E(n, m) \end{aligned}$$

   a) Define a natural map $n = m \to E(n, m)$ for all $n, m : \mathbb{N}$.
   b) Use the fundamental theorem of identity types to show this natural map is an equivalence.

10. Show that a type X is $n+1$-truncated if and only if the diagonal map of X,

$$\delta_X : X \to X \times X, \quad x \mapsto (x, x),$$

is n-truncated.

11. Show that $\|X\|_0$ satisfies the universal property of the set quotient of X modulo the equivalence relation $\sim$, where $x \sim x'$ if and only if $\|x =_X x'\|$.

1.2 Solutions

1. Define $f : \|\,\|A\|_n\|_k \to \|A\|_k$ to be $f(\|a|_n|_k) = |a|_k$ and $g : \|A\|_k \to \|\,\|A\|_n\|_k$ to be $g(|a|_k) = \|a|_n|_k$. Both functions are defined using the universal property of truncations, and the fact that $\|A\|_k$ is n-truncated because $k \leq n$. It's then easy to see that f and g define a quasi-eqivalent.

2. First assume X is an n-connected type. Then $\|X\|_n$ is contractible, and hence for every n-type Y, we apply the universal property of n-truncation and get:

$$(X \to Y) \simeq (\|X\|_n \to Y) \simeq Y$$

Therefore, we can identify δ with id_Y, and so it is an equivalence.

Conversely, let $\delta_Y : Y \to (X \to Y)$ defines an equivalence for all n-types Y. So in particular it is an equivalence for $Y := \|X\|_n$. Applying this on $|\text{-}|_n : X \to \|X\|_n$, we get:

$$\mathrm{fib}_{\delta_{\|X\|_n}} |\text{-}|_n$$

is contractible. Unfolding this, we get that the type

$$(z : \|X\|_n) \times (\mathsf{const}_z = |\text{-}|_n)$$

is contractible. By function extensionality, we get that

$$(z : \|X\|_n) \times ((x : X) \to |x|_n = z)$$

is contractible. Now since identities of $\|X\|_n$ are $(n-1)$-truncated, and so n-truncated, we can apply universal property of truncation to get that $\|X\|_n$ is contractible.

3. First, show that the first projection map from $(z : \|A\|_n) \times \|\,\mathrm{fib}_{|\text{-}|_n} z\|_n$ to $\|A\|_n$ is an equivalence, using that the domain is an n-type. Then apply the following lemma:

   Lemma. *For any type A and a family of types $B : A \to \mathcal{U}$, the total space $(x : A) \times B(x)$ is contractible if and only of for each $x : A$, $B(x)$ is a contractible type.*

4. The map i is clearly n-truncated as its fibers are n-truncation and therefore n-types. We can identify X with the total space of fibers $(y : Y) \to \mathrm{fib}_f(y)$. Then p is n-connected by observing that the map between total spaces is n-connected if maps between fibers are, and the later is the result of the previous exercise.

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8. Since X is a connected type, we have a map f assigns to each two points $x_1, x_2 : X$ a path $f(x_1, x_2) : |x_1|_0 =_{\|X\|_0} |x_2|_0$. On the other hand, we know that any path $p : x_1 = x_2$ implies the goal, and since the goal is a proposition, it is sufficient to find a proof for $\|x_1 = x_2\|$, which can be found by applying the following lemma.

   Lemma. *For any type X and any $x_1, x_2 : X$, there is an equivalence*

   $$\big(\, |x_1|_0 =_{\|X\|_0} |x_2|_0 \,\big) \;\simeq\; \|\, x_1 =_X x_2 \,\|.$$

   Proof. Since $\|X\|_0$ is a set, the type

   $$|x_1|_0 =_{\|X\|_0} |x_2|_0$$

is a proposition; and $\|x_1 = x_2\|$ is also a proposition. Hence it suffices to give maps in both directions.

For

$$\|x_1 = x_2\| \to \big(|x_1|_0 = |x_2|_0\big),$$

use propositional truncation recursion and send a path

$$p : x_1 = x_2$$

to

$$\mathrm{ap}_{|-|_0}(p) : |x_1|_0 = |x_2|_0.$$

For the converse, define the proposition-valued family

$$P : \|X\|_0 \to \mathcal{U}, \qquad P(z) :\equiv \|\, x_1 = z \,\|,$$

where P is induced from the family $y \mapsto \|x_1 = y\|$ on X by set-truncation recursion. Then $P(|x_1|_0)$ is inhabited by $|\mathrm{refl}_{x_1}|$. Thus, given

$$q : |x_1|_0 = |x_2|_0,$$

transporting along q yields an inhabitant of

$$P(|x_2|_0) \equiv \|x_1 = x_2\|.$$

Therefore the two proposition-valued types are equivalent. Note that one can generalize this result to n-truncation (see the HoTT book, Theorem 7.3.12). □

9.

10. For each $(x_1, x_2) : X \times X$, we have:

$$\mathrm{fib}_{\delta_X}(x_1, x_2) = (x : X)\times((x, x) = (x_1, x_2)) = (x : X)\times(x = x_1 \times x = x_2) = x_1 = x_2,$$

and we know that X is $(n+1)$-truncated if and only if its identity types are n-truncated.

11. The relation

$$x \sim x' \;\equiv\; \|x =_X x'\|$$

is an equivalence relation: reflexivity is given by $|\mathrm{refl}_x|$, while symmetry and transitivity follow by truncation elimination into propositions.

Now let A be a set. To show that $\|X\|_0$ is the set quotient of X by $\sim$, we must show that precomposition with the truncation map

$$|-|_0 : X \to \|X\|_0$$

induces an equivalence

$$(\|X\|_0 \to A) \;\simeq\; \Big\{ f : X \to A \mid \forall x, x' : X,\; x \sim x' \to f(x) = f(x') \Big\}.$$

Given $g : \|X\|_0 \to A$, define $f : X \to A$ by $f(x) := g(|x|_0)$. Since A is a set, the type $f(x) = f(x')$ is a proposition. Hence by truncation elimination it suffices to consider $p : x = x'$, in which case

$$f(x) = g(|x|_0) = g(|x'|_0) = f(x')$$

by applying g to $\mathrm{ap}_{|-|_0}(p)$.

Conversely, suppose $f : X \to A$ respects $\sim$. Since A is a set, set-truncation recursion yields a map

$$\bar{f} : \|X\|_0 \to A, \qquad \bar{f}(|x|_0) := f(x),$$

and this is well-defined exactly because f respects $\sim$.

These two constructions are inverse to each other, by the computation rule for set truncation and by set-truncation induction. Therefore $\|X\|_0$ satisfies the universal property of the set quotient of X modulo $\sim$.

Chapter 2

A menagerie of higher inductive types

2.1 Exercises

1. (Pushout-Pullback Duality) Consider a commutating square:

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

   with $H : i \circ f \sim j \circ g$. Then the following are equivalent:

   (i) The square is a pushout square.

   (ii) the square

$$\begin{tikzcd} T^X \arrow[r, "-\circ j"] \arrow[d, "-\circ i"'] & T^C \arrow[d, "-\circ g"] \\ T^B \arrow[r, "-\circ f"'] & T^A \end{tikzcd}$$

   is a pullback square, for every type T.

2. Show that the two definitions of S^1 are equivalent, in two ways:

   (i) Show the HIT for S^1 is the same thing as $\Sigma 2$ by constructing an equivalence between them.

   (ii) By applying Exercise 2.1.

3. Show that evaluation at the loop gives pointed equivalence

$$(S^1 \to_\star X) \to_\star \Omega X$$

for any pointed type X.

Conclude, using the suspension–loop adjunction, that we have a natural equivalence

$$(S^n \to_\star X) \to_\star \Omega^n X$$

for all $n : \mathbb{N}$.

4. Show that a type X is a proposition iff $\mathrm{inl} : X \to X * X$ is an equivalence.

5. Show that $A \vee B \simeq A * B$ for propositions A, B, where $A \vee B := \|A + B\|$. (Beware the notational overload between the wedge sum and disjunction: It's not too bad in practice, because the only pointed proposition is the unit type!)

6. (i) Consider a span $A \xleftarrow{f} X \xrightarrow{g} B$. Show that the following square is pushout:

$$\begin{tikzcd} X + X \arrow[r, "{[\mathrm{id},\mathrm{id}]}"] \arrow[d, "f+g"'] \arrow[dr, phantom, "\ulcorner"] & X \arrow[d, "inr \circ g"] \\ A + B \arrow[r, "{[\mathrm{inl},\mathrm{inr}]}"'] & A \amalg_X B \end{tikzcd}$$

   (ii) Apply this to show that for every type X we get $\Sigma X \simeq \mathsf{bool} * X$.

   . (Harder) Show that $S^1 \wedge X \simeq \Sigma X$

7. (Smash Product - Pointed Map Space Adjunction)

   (i) Prove that for each pointed type B, the fucnction $(-) \wedge B$ is the left adjoint to the function $B \to_* (-)$,i.e. for every pointed types A and C, there is a pointed equivalence:

$$A \to_* (B \to_* C) \simeq_* (A \wedge B) \to_* C$$

   (ii) Use this to prove that $S^1 \wedge X \simeq \Sigma X$ for each type X.

8. Show that the smash product is equivalently the pushout:

$$\begin{tikzcd} X + Y \arrow[r] \arrow[d] & X \times Y \arrow[d] \\ 1 + 1 \arrow[r] & X \wedge Y \end{tikzcd}$$

9. (i) For all pointed types X, Y, and A, prove there is a pointed equivalence:

$$X * Y \to_* A \simeq_* X \to (Y \to_* \Omega A)$$

   (ii) For pointed types X, Y prove that

$$X * Y \simeq_* \Sigma(X \wedge Y).$$

10. We can define the unit interval type $\mathbb{I}$ to be the pushout of the constant span $1 \leftarrow 1 \rightarrow 1$. Working in Martin-Löf type theory *without* univalence (and function extensionality) but with pushouts with strict definitional computation rules, prove that

    (i) $[((x, y) : A \times A) \times (x = y)] \simeq (\mathbb{I} \rightarrow A)$

    (ii) Use the above equivalence (or, if you prefer, a variation thereof) to derive function extensionality.

11. (i) Show that if the square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

    is a pullback, then so is

$$\begin{tikzcd} \Omega A \arrow[r, "\Omega g"] \arrow[d, "\Omega f"'] & \Omega C \arrow[d, "\Omega j"] \\ \Omega B \arrow[r, "\Omega i"'] & \Omega X \end{tikzcd}$$

    (ii) For a map $A \xrightarrow{f} B$ show that $\Omega\,(\mathrm{fib}\ f) \simeq \mathrm{fib}\,(\Omega f)$

12. Show that the square

$$\begin{tikzcd} S^1 \arrow[r] \arrow[d] & 1 \arrow[d] \\ S^1 \times S^1 \arrow[r] & S^2 \vee S^1 \end{tikzcd}$$

    is a pushout.

2.2 Solutions

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10. Write the interval as the pushout of the constant span $1 \leftarrow 1 \rightarrow 1$, with points $0, 1 : \mathbb{I}$ and path $\mathsf{seg} : 0 = 1$.

    (i) Define
    $$\Phi : \Big(\sum_{x,y:A} (x = y)\Big) \to (\mathbb{I} \to A)$$
    by interval recursion:
    $$\Phi(x, y, p) \;:=\; \mathsf{rec}_{\mathbb{I}}(x, y, p).$$
    Thus, by the strict β-rules, we get:
    $$\Phi(x, y, p)(0) \equiv x, \qquad \Phi(x, y, p)(1) \equiv y, \qquad \mathrm{ap}_{\Phi(x,y,p)}(\mathsf{seg}) \equiv p.$$
    Conversely, define
    $$\Psi : (\mathbb{I} \to A) \to \sum_{x,y:A} (x = y)$$
    by
    $$\Psi(h) \;:=\; \big(h(0),\ h(1),\ \mathrm{ap}_h(\mathsf{seg})\big).$$
    Then
    $$\Psi(\Phi(x, y, p)) \equiv (x, y, p)$$
    judgmentally, by the strict β-rules above. For the other composite, if $h : \mathbb{I} \to A$, then
    $$\Phi(\Psi(h)) = \Phi\big(h(0), h(1), \mathrm{ap}_h(\mathsf{seg})\big)$$
    is the map out of $\mathbb{I}$ with the same values on $0, 1$ and the same action on seg as h; hence, by the η-rule of the pushout $\mathbb{I}$,
    $$\Phi(\Psi(h)) = h.$$
    Therefore
    $$\Big(\sum_{x,y:A} (x = y)\Big) \simeq (\mathbb{I} \to A).$$
    Note that we do not actually need strict β-rule for this part.

(ii) First we derive non-dependent function extensionality. Let $f, g : A \to B$ and assume

$$H : \prod_{a:A} (f(a) = g(a)).$$

By part (i), for each $a : A$ we obtain a map

$$\widetilde{H}(a) : \mathbb{I} \to B$$

such that

$$\widetilde{H}(a)(0) \equiv f(a), \qquad \widetilde{H}(a)(1) \equiv g(a), \qquad \mathrm{ap}_{\widetilde{H}(a)}(\mathsf{seg}) \equiv H(a).$$

Now curry these maps:

$$K : \mathbb{I} \to (A \to B), \qquad K(i)(a) := \widetilde{H}(a)(i).$$

Then, using the strict computation rules from part (i) and the judgmental η-rule for functions,

$$K(0) \equiv \lambda a.\ \widetilde{H}(a)(0) \equiv \lambda a.\ f(a) \equiv f,$$

and similarly

$$K(1) \equiv g.$$

Hence

$$\mathrm{ap}_K(\mathsf{seg}) : f = g.$$

So ordinary function extensionality holds.
To obtain dependent function extensionality, let $B : A \to \mathcal{U}$, let

$$f, g : \prod_{a:A} B(a), \qquad H : \prod_{a:A} (f(a) = g(a)).$$

Define

$$f', g' : A \to \sum_{a:A} B(a), \qquad f'(a) := (a, f(a)), \quad g'(a) := (a, g(a)).$$

For each $a : A$, using $H(a)$ we get

$$f'(a) = g'(a)$$

in the Σ-type, namely by the path $(\mathrm{refl}_a, H(a))$. By ordinary function extensionality, this yields

$$p : f' = g'.$$

Now apply ap to the map

$$s : \left(A \to \sum_{a:A} B(a)\right) \to \prod_{a:A} B(a), \qquad s(k)(a) := \pi_2(k(a)).$$

Since $s(f') = f$ and $s(g') = g$, we obtain

$$\mathrm{ap}_s(p) : f = g.$$

Thus full function extensionality follows. It seems that we can prove (ii) without assuming η-rule for $\mathbb{I}$, right? Since we only need the right to left map from (i), and not the whole equivalence.

11. (i) Assume the given square is a pullback. Then $A \simeq B \times_X C$. Since for any pointed type Y we have

$$\Omega Y \simeq 1 \times_Y 1,$$

and pullbacks commute with pullbacks (by 3-3 lemma for pullbacks), it follows that

$$\Omega A \;\simeq\; \Omega(B \times_X C) \;\simeq\; (1 \times_B 1) \times_{(1 \times_X 1)} (1 \times_C 1) \;\simeq\; \Omega B \times_{\Omega X} \Omega C.$$

(ii) The homotopy fiber of f is the pullback

$$\mathrm{fib}(f) \simeq 1 \times_B A.$$

Apply (i) to this pullback square.

12. Using the equation $S^1 \wedge X \simeq \Sigma X$ we can identify S^2 with $S^1 \wedge S^1$, which is pushout of $S^1 \vee S^1 \to S^1 \times S^1$ along the unit map. Now we have:

$$\begin{tikzcd}
S^1 \vee S^1 \arrow[r] \arrow[d] & 1 \arrow[r] \arrow[d] & S^1 \arrow[d] \\
S^1 \times S^1 \arrow[r] & S^2 \arrow[r] & S^2 \vee S^1
\end{tikzcd}$$

Since both squares are pushouts, the pasting lemma for pushout says that the outer rectangle is a pushout. We get the goal by observing that the upper square in the following diagram is a pushout:

$$\begin{tikzcd}
S^1 \arrow[r] \arrow[d] & 1 \arrow[d] \\
S^1 \vee S^1 \arrow[r] \arrow[d] & S^1 \arrow[d] \\
S^1 \times S^1 \arrow[r] & S^2 \vee S^1
\end{tikzcd}$$

Chapter 3

The Hopf fibration and the long exact sequence

3.1 Exercises

1. Construct the product on the circle directly by double circle induction.

2. Give another proof of **??** by showing that $\mathrm{ap}_\varphi : (* = *) \to (\varphi(*) = \varphi(*))$ is an equivalence.

3. Construct a retraction $\rho : \Omega S^2 \to S^1$ of $\eta : S^1 \to \Omega S^2$. (*Hint*) Use the Hopf fibration.

4. We can define a "tensor product" of $\mathbb{Z}$-torsors by setting

$$(X, t) \otimes (Y, u) := (X \times Y/\sim, s)$$

   using the set quotient by the equivalence relation generated by $(t(x), y) \sim (x, u(y))$, and setting $s[(x, y)] := [t(x), y)]$. Check that this operation is well defined and gives an H-space structure on $\mathrm{Tors}_{\mathbb{Z}}$ with the base point $(\mathbb{Z}, \mathrm{suc})$ as neutral element.

5. Investigate some other components of $\mathrm{Set}_{\circlearrowleft}$, in particular those containing $(\mathrm{Fin}\, n, \mathrm{suc})$, where $\mathrm{Fin}\, n = (k : \mathbb{N}) \times (k < n)$ and the successor operation is taken modulo n. If this component is denoted Cyc_n (for n-cycles), what are the maps $\mathrm{Cyc}_n \to \mathrm{Cyc}_m$?

Chapter 4

The Freudenthal Suspension Theorem and Degrees

4.1 Exercises

1. Deduce the dual Blakers–Massey **??** from the Join Connectivity **??** by identifying the fibers of the cogap map with the joins of the fibers of legs in the cospan.

2. Deduce the Wedge Connectivity Lemma from the dual Blakers–Massey **??**, i.e., consider the pullback square, for pointed types X, Y,

$$\begin{tikzcd} 1 \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d] \\ X \arrow[r] & X \times Y. \end{tikzcd}$$

   Show that if X is n-connected and Y is m-connected, then the wedge inclusion map $X \vee Y \to X \times Y$ is $(n+m)$-connected.

3. Show that the counit $\varepsilon : \Sigma\Omega X \to X$ of the loop–suspension adjunction is $2n$-connected if X is n-connected.

4. Show that the pushout of an embedding is an embedding, i.e., if g is (-1)-truncated, then so is inl in the pushout square:

$$\begin{tikzcd} R \arrow[r, "g"] \arrow[d, "f"'] & B \arrow[d, "\mathrm{inr}"] \\ A \arrow[r, "\mathrm{inl}"'] \arrow[ur, phantom, "\ulcorner", very near start] & Q \end{tikzcd}$$